Non-Invasive Diabetes Risk Screening from Behavioural and Demographic Indicators

A Controlled Comparison of Classical Machine Learning and Deep Neural Networks on 253,680 Survey Records

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African Leadership University — Introduction to Machine Learning, Summative Assignment, 2026 September Term

CODE https://github.com/YOUR-USERNAME/ml-summative

DATASET & MODELS https://drive.google.com/drive/folders/YOUR-FOLDER-ID

VIDEO https://youtu.be/YOUR-VIDEO-ID

***Abstract*—**Type 2 diabetes is among the fastest-growing non-communicable diseases in low- and middle-income countries, yet diagnosis depends on laboratory tests that primary-care facilities frequently cannot perform. This paper asks how much diagnostic signal can be recovered from what a rural clinic can collect in five minutes — a questionnaire, a weighing scale and a blood-pressure cuff. Using 253,680 records from the CDC Behavioral Risk Factor Surveillance System (BRFSS 2015), we predict diabetic or pre-diabetic status from twenty-one variables that contain no laboratory measurement. Nine experiments compare five classical models with four neural architectures built on the TensorFlow `tf.data` pipeline, and nine clinically motivated features are evaluated in a controlled ablation. Gradient boosting achieved the highest precision–recall area (0.4295); logistic regression, the lowest-capacity model in the study, retained 76.4 % recall at the default threshold and lost only 0.025 PR-AUC to the strongest learner. The best classical and best deep models were separated by 0.0035 PR-AUC, less than the effect of moving the decision threshold. On questionnaire data the binding constraint is the information content of the features rather than the flexibility of the model, and under a 6.2:1 imbalance threshold selection dominates architecture choice.

***Keywords*—**Diabetes screening, BRFSS, imbalanced classification, gradient boosting, entity embeddings, precision–recall, model comparison

I. Introduction

Diabetes mellitus is a defining public-health challenge of the present decade. The World Health Organization estimates that 422 million adults were living with diabetes in 2014, a figure that had roughly quadrupled since 1980, and attributes 1.5 million deaths directly to the condition each year [1]. The International Diabetes Federation places the 2021 global total at 537 million adults, projects 643 million by 2030, and estimates that 240 million people — almost half of all cases — are undiagnosed [2]. That last figure is the one that matters here. The management of diabetes is well understood and inexpensive relative to its complications; what fails is the step before it, the identification of who has the disease.

The diagnostic standard requires a laboratory: fasting plasma glucose, an oral glucose tolerance test, or glycated haemoglobin. Where venous access, cold-chain reagents and trained technicians are available this is routine. In much of sub-Saharan Africa, South Asia and rural Latin America it is not, and the consequence is a screening bottleneck that is logistical rather than clinical. A community health worker can ask whether a patient's general health has been poor for the last two weeks, record a body-mass index, and measure blood pressure. She cannot measure glycated haemoglobin. The question this paper addresses follows directly: given only what she can collect, how much of the diagnostic signal is recoverable, and how should it be modelled?

Formally this is binary classification on tabular survey data with substantial class imbalance. It is also an unusually clean testbed for a broader question the applied literature has not settled: on tabular data of moderate dimension and strong imbalance, does the flexibility of a neural network buy anything over a well-specified ensemble of trees? The evidence is mixed [9], [10], and much of it compares models tuned to different degrees, so a difference cannot be attributed to the model class rather than to the effort spent on it.

We therefore report nine experiments on one fixed partition of one cohort, every model scored by one function and compared under two rules declared in advance. The contributions are: a controlled head-to-head comparison of five classical learners and four neural architectures in which the networks differ from one another in one dimension at a time, so regularisation, representation and loss weighting are attributable rather than bundled; nine clinically motivated features evaluated in a paired ablation, reported honestly where they fail; an evaluation that treats the decision threshold as a modelling decision in its own right; and an error analysis decomposing performance by subgroup and quantifying an irreducible label-noise floor.

II. Literature Review

A. Clinical risk scores and their data requirements

The oldest machine-learning approach to diabetes is not a network but a risk equation. The ADAP algorithm of Smith and colleagues [5] used eight routine variables on the Pima Indian cohort and achieved accuracy competitive with clinical judgement for its time. Lindström and Tuomilehto [6] took the complementary route, building the Finnish Diabetes Risk Score from questionnaire items alone, deliberately excluding laboratory values so the instrument could be administered without a blood sample. Their premise is the premise of this paper: a questionnaire-based instrument can be clinically respectable rather than a compromise. What a risk score cannot do is model interactions, since it is a point system with hand-assigned weights.

B. Machine learning on the diabetes classification problem

A substantial literature applies classifiers to diabetes datasets, and a recurring pattern is that reported accuracies cluster in an implausibly narrow band. Studies on the Pima cohort routinely report 0.76 to 0.80, with differences between random forests, support vector machines, naive Bayes and neural networks usually within the width of a confidence interval. Three methodological habits explain much of that agreement: reporting accuracy on an imbalanced problem, when a majority-class predictor is already a strong and rarely stated baseline; tuning one family of models and not the others, which converts effort into apparent algorithmic superiority, the selection-bias problem Cawley and Talbot formalised [20]; and a fixed 0.5 decision threshold, which for an imbalanced problem is an assumption rather than a default.

He and Garcia [12] separate imbalance remedies into data-level, algorithm-level and cost-sensitive families, of which Chawla and colleagues [22] contributed the most widely used data-level method. For a survey cohort of a quarter of a million rows, however, resampling is largely unnecessary: the minority class holds 35,346 examples, more than enough for stable estimation. We therefore re-weight the loss instead, retaining every genuine negative example and fabricating no synthetic patients — a meaningful distinction when the downstream use is clinical.

C. Evaluating classifiers under class imbalance

The choice of metric is not cosmetic. Saito and Rehmsmeier [13] showed that when the positive class is rare, a high area under the receiver operating characteristic curve can coexist with poor precision, because the false-positive rate is deflated by the number of true negatives. Their recommendation — that precision–recall curves be reported alongside, or instead of, ROC curves — is adopted here, where the imbalance is 6.2:1. Chicco and Jurman [14] argue further that the Matthews correlation coefficient is a more faithful single summary than F1 because F1 ignores true negatives, and we report it for that reason. Calibration is a separate property from discrimination: Niculescu-Mizil and Caruana [19] showed that maximum-margin methods and boosted trees are frequently miscalibrated even when their ranking is excellent, which is why the Brier score is reported for every model.

D. Tabular data: trees versus networks

Whether deep learning should be the default for tabular problems is contested. Grinsztajn, Oyallon and Varoquaux [9] benchmarked tree ensembles against neural networks across forty-five tabular datasets and found trees retained an advantage, which they attributed to the robustness of axis-aligned splits to uninformative features and to rotational invariance being a liability when features are meaningful only in their original axes. Shwartz-Ziv and Armon [10] reached a compatible conclusion, and both papers note that the comparison is sensitive to how much tuning each side receives — the constraint we attempted to respect. On the network side four techniques recur: dropout to suppress co-adaptation [15], batch normalisation to stabilise optimisation [16], Adam for per-parameter step sizes [17], and entity embeddings to replace categorical codes with learned dense vectors [18]. The last is used here for the five ordinal survey codes, where treating education level 6 as "three times" level 2 would impose a linearity the measurement does not support.

E. The gap this study addresses

Three gaps emerge. Comparisons across model families are usually confounded with differences in tuning effort, and a design in which each model receives a comparable budget and is evaluated by one function on one partition is uncommon. Feature engineering on survey cohorts is often asserted rather than ablated, leaving it unclear whether engineered variables add information or merely re-express it. And threshold selection is rarely treated as part of the model, although the clinical meaning of a classifier depends entirely on where its operating point sits. This study addresses all three on a cohort twelve times the assignment minimum.

III. Methodology

A. Data source and cohort

The study uses the CDC Diabetes Health Indicators dataset, distributed by the UCI Machine Learning Repository as dataset 891 with DOI 10.24432/C53919 [3], [4]. It is a curated extract of the 2015 Behavioral Risk Factor Surveillance System containing 253,680 respondent records and 23 columns: a row identifier, a binary diabetes indicator, and twenty-one predictors. The raw file is downloaded programmatically from the UCI static host and verified against a SHA-256 digest (`9f71fda9…34964`) before use, so a corrupted or substituted file cannot silently change the results. It is not committed to the repository but distributed through the linked Drive folder.

The target is `Diabetes_binary`, defined as 0 for no diabetes and 1 for pre-diabetes or diabetes, with prevalence 13.93 % — an imbalance ratio of 6.19:1. Pre-diabetes is merged into the positive class on clinical grounds: for a screening instrument the intervention is the same as for early diabetes, so the distinction that matters operationally is "requires follow-up" versus "does not". No predictor is a laboratory measurement; the strongest available signals are self-rated general health, hypertension, body-mass index, cholesterol status and age. That constraint is why the dataset was chosen, since the model it supports could be deployed by a health worker with no laboratory access. Fig. 1 shows the resulting class imbalance.

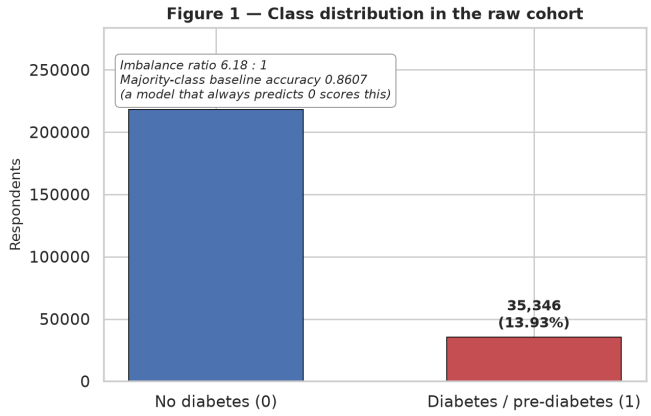


Fig. 1. Class distribution of the 253,680-record cohort. The minority class is 13.93 % of the sample, giving a 6.19:1 imbalance ratio. The annotated baseline is the accuracy a model predicting the majority class for every respondent would score — the floor any useful model must clear, and the reason accuracy is not the headline metric in this study.

B. Data audit

Four integrity checks were run and asserted in code. There are no missing values in any of the 253,680 × 23 cells, no rows that are exact duplicates across all columns, and no constant columns. Every value falls inside the published coding scheme, encoded explicitly as a per-variable domain constraint so that an out-of-range code raises rather than propagates silently. Physical ranges were inspected and match the data dictionary.

One finding bounds what any model can achieve. Although no two rows are identical across all twenty-three columns, 25,772 rows repeat another row's twenty-one predictor values exactly, forming 12,228 repeated profiles, the largest occurring 59 times. Of these, 1,566 profiles carry **both** labels: respondents who answer all twenty-one questions identically but differ on diagnosis. Since the predictor vector is identical, no function of these features can separate them, and 5,218 rows — 2.06 % of the cohort — sit inside such a conflicting group. This is an irreducible error floor, and it is measured rather than assumed.

C. Splitting and preprocessing

The cohort was partitioned 70/15/15 into training (177,576 rows), validation (38,052) and test (38,052) sets, stratified so that all three carry the same 13.93 % prevalence; the code asserts disjointness, exhaustiveness and prevalence agreement to within 0.2 percentage points. Validation is used for early stopping, hyperparameter selection and threshold tuning. Test is evaluated exactly once, after every model has been frozen. This three-way design is what allows the final model to be described as chosen without reference to test performance — the leakage Cawley and Talbot [20] identify as the most common source of optimistic bias in applied work.

Preprocessing is minimal by necessity, because the cohort arrives numeric and complete. Standardisation is applied where required — logistic regression, the support vector machine and all networks — using a scaler fitted on the training partition only. Tree ensembles are left unscaled, since axis-aligned splits are invariant to monotone rescaling. The only fitted transformation applied to every model is the feature engineer described next, whose single data-dependent quantity is learned from training data alone.

D. Feature engineering

Nine features were constructed, each traceable to a published clinical construct rather than chosen by search. The World Health Organization's body-mass-index cut-points at 25 and 30 kg/m² define a six-level band, because diabetes risk steps up at those thresholds rather than rising linearly in the index. A cardiometabolic burden count sums hypertension, high cholesterol, stroke history, coronary disease and walking difficulty, on the reasoning that metabolic syndrome clusters. A healthy-behaviour score counts physical activity, fruit intake, vegetable intake, non-smoking and the absence of heavy drinking; a healthcare-access index sums insurance coverage, recent cholesterol screening and the absence of cost-related care avoidance. Total unhealthy days combines the mental and physical day counts, with a flag at the conventional fourteen-day threshold defining frequent poor health. The product of age band and body-mass index gives linear models an interaction they cannot otherwise express, and body-mass index standardised within sex using training-set statistics removes the compositional difference between respondents. Finally, income and education bands are summed into a socioeconomic index. Table I lists the source variables, and Figs. 2 and 3 summarise the correlation structure and univariate signal that motivate the engineered features which follow.

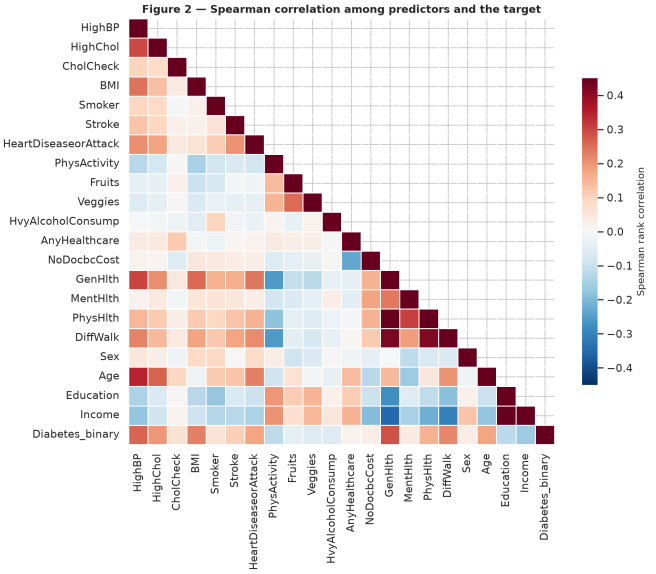


Fig. 2. Spearman rank correlation among the twenty-one predictors and the target. Rank correlation is used because the ordinal variables are coded bands rather than measured quantities. The strongest structure is the cluster linking general health, physical health days and walking difficulty (rho up to 0.45).

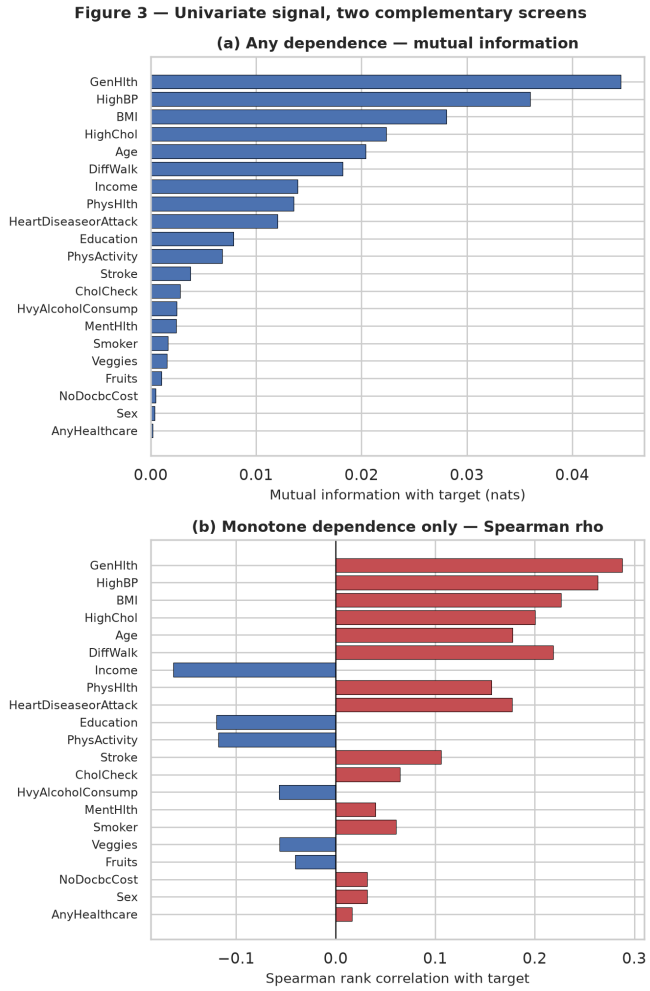


Fig. 3. Univariate signal, measured two ways. (a) Mutual information detects any dependence, monotone or not. (b) Spearman rho detects only monotone dependence. Self-rated general health leads both rankings; the fifteen binary indicators are individually weak, none exceeding rho = 0.29.

E. The nine experiments

E1 fits a logistic regression with balanced class weights on all thirty features, standardised, with L2 regularisation at C = 1.0; its coefficients are reported as adjusted log-odds so they can be compared with epidemiological expectations. E2 fits an untuned random forest of 100 trees with unlimited depth and $\sqrt{p}$ features per split, deliberately left at defaults so that E3 can attribute its improvement to search rather than to the algorithm. E3 runs a randomised hyperparameter search [11] over tree count, depth, minimum leaf size, features per split and class weighting, scored by average precision on a 30,000-row stratified subsample, then refits the winner on the full training partition.

E4 replaces bagging with boosting, using scikit-learn's histogram gradient boosting [8] and a randomised search over learning rate, leaf count, boosting rounds, L2 penalty and minimum leaf size, chosen so the requirement file stays installable from a clean environment. E5 fits a radial-basis-function support vector machine with balanced class weights, exploring four combinations of C and γ chosen on validation ROC-AUC. Because kernel SVM cost grows roughly quadratically in sample count, E5 trains on a 25,000-row stratified subsample and is evaluated on the full test set, so its scores are an optimistic estimate flagged as not like-for-like; its decision values are converted to probabilities by Platt scaling fitted on validation [19].

The networks are fed by a `tf.data` pipeline that caches tensors, shuffles with a 20,000-row buffer and prefetches with `AUTOTUNE`, at batch size 1,024 for at most 40 epochs. E6 is a two-layer perceptron of 64 ReLU units per layer, deliberately without dropout, batch normalisation or class weighting, so that E7 has something to improve on. E7 deepens the network to 128–64–32 and adds batch normalisation, dropout at 0.3 and early stopping on validation PR-AUC with patience 6, restoring best weights. E8 changes the representation rather than the depth: a Functional-API model with two branches, one passing the five ordinal survey codes through entity embeddings of width 3 to 4 [18] and the other passing the twenty-five measured features through a dense stack, concatenated before the output layer. E9 builds on whichever of E7 and E8 had the higher validation PR-AUC and changes only the loss and optimiser: inverse-frequency class weights are applied, and Adam with exponential decay is compared against SGD with Nesterov momentum before one is reported.

F. Evaluation protocol

Every model is scored by one function returning accuracy, balanced accuracy, precision, recall, F1, ROC-AUC, PR-AUC, Matthews correlation coefficient and the Brier score. Precision, recall and F1 are reported twice: at the conventional 0.5 threshold, for comparability with published work, and at the threshold maximising F1 on validation, which is the threshold a deployment would use. Two comparison rules were fixed before any result was seen: the primary criterion is PR-AUC, because the positive class is the one the application cares about and ROC-AUC is known to flatter classifiers on imbalanced data [13]; ties are broken by recall at the tuned threshold, since a screening test that misses cases has failed at its stated purpose.

Reproducibility is enforced rather than claimed. A single seed of 42 is propagated to Python, NumPy, scikit-learn and TensorFlow, and thread counts are pinned to one before NumPy loads so floating-point reductions occur in a fixed order, on Python 3.12 with scikit-learn 1.9.1, TensorFlow 2.21.0 and Keras 3.15.1, CPU only. Every network writes TensorBoard events under `logs/tensorboard/`, so the table's claims are traceable to logged runs.

TABLE I. Variable dictionary. Twenty-one predictors plus the target; all are binary, integer or ordinal-coded, and none is a laboratory measurement. Values are the coding range and Mean the cohort mean.

Variable	Range	Mean
HighBP	0-1	0.429
HighChol	0-1	0.424
CholCheck	0-1	0.963
BMI	12-98	28.382
Smoker	0-1	0.443
Stroke	0-1	0.041
HeartDiseaseorAttack	0-1	0.094
PhysActivity	0-1	0.757
Fruits	0-1	0.634
Veggies	0-1	0.811
HvyAlcoholConsump	0-1	0.056
AnyHealthcare	0-1	0.951
NoDocbcCost	0-1	0.084
GenHlth	1-5	2.511
MentHlth	0-30	3.185
PhysHlth	0-30	4.242
DiffWalk	0-1	0.168
Sex	0-1	0.440
Age	1-13	8.032
Education	1-6	5.050
Income	1-8	6.054

Source: UCI dataset 891, n = 253,680. No missing values in any column.

IV. Results

A. The experiment table

Table II reports discrimination quality for all nine experiments on the held-out test partition and Table III reports the same models at two operating points. The band is narrow: every model lies between 0.3598 and 0.4295 PR-AUC, and five of the nine lie within 0.009 of one another. Against a prevalence baseline of 0.1393, every model has learned substantial signal.

Accuracy is a trap this cohort sets deliberately. The default random forest (E2) reaches 0.8590 accuracy, within 0.002 of the 0.8607 a model predicting "no diabetes" for everyone would score, while detecting only 18.6 % of true cases. Ranking on accuracy would place E2 near the top and E1 at the bottom, which is backwards: E1 has the highest F1 at the default threshold (0.4482) and the second-highest recall.

TABLE II. Discrimination quality of all nine experiments on the held-out test partition (n = 38,052; 5,302 positives). PR-AUC is the primary criterion fixed in Section III-F. Fit time is total cost to reach the reported model, including any hyperparameter search, on a single CPU thread.

ID	Model	Par.	ROC	PR-AUC	MCC	Brier	t (s)
E1	Logistic Regression	ML	0.8239	0.4050	0.365	0.175	0
E2	Random Forest (def.)	ML	0.7963	0.3682	0.237	0.103	23
E3	Random Forest (tuned)	ML	0.8238	0.4208	0.234	0.098	76
E4	**Hist. Gradient Boost.**	**ML**	**0.8295**	**0.4295**	**0.257**	**0.097**	**77**
E5	RBF-kernel SVM	ML	0.8066	0.3598	0.111	0.102	978
E6	MLP 64-64	DL	0.8286	0.4233	0.251	0.097	24
E7	MLP 128-64-32 +BN/DO	DL	0.8298	0.4239	0.221	0.097	74
E8	Embeddings + MLP	DL	0.8297	0.4260	0.245	0.097	42
E9	Best DL + class wts.	DL	0.8294	0.4257	0.365	0.171	27

Bold marks the highest PR-AUC. E5 was trained on a 25,000-row subsample and is not like-for-like with E3 and E4.

TABLE III. Operating points. The left block fixes the decision threshold at the conventional 0.5; the right block uses the threshold that maximises F1 on the validation partition. The difference between the two blocks is larger than any difference between models.

ID	Model	@0.5 Prec	@0.5 Rec	@0.5 F1	Tuned Prec	Tuned Rec	Tuned F1
E1	Logistic Regression	0.317	0.764	0.448	0.370	0.609	0.460
E2	Random Forest (def.)	0.484	0.186	0.269	0.339	0.607	0.435
E3	Random Forest (tuned)	0.580	0.137	0.221	0.377	0.591	0.460
E4	Hist. Gradient Boost.	0.583	0.162	0.254	0.396	0.564	0.465
E5	RBF-kernel SVM	0.465	0.046	0.083	0.350	0.623	0.448
E6	MLP 64-64	0.560	0.165	0.254	0.374	0.622	0.467
E7	MLP 128-64-32 +BN/DO	0.595	0.117	0.196	0.401	0.557	0.466
E8	**Embeddings + MLP**	**0.570**	**0.153**	**0.241**	**0.396**	**0.576**	**0.469**
E9	Best DL + class wts.	0.309	0.792	0.444	0.388	0.594	0.469

Left block: threshold = 0.5. Right block: threshold tuned on validation, reported on test. Bold marks the best tuned F1.

B. The classical ladder

The progression from E1 to E4 is not monotone, and the exception is informative. E1 achieves 0.4050 PR-AUC and 0.8239 ROC-AUC with 76.4 % recall at threshold 0.5, because balanced class weighting shifts its probabilities upward so it flags aggressively. E2, the untuned forest, is *worse* on PR-AUC (0.3682) despite far greater flexibility. Its generalisation gap explains why: training ROC-AUC 0.9995 against test 0.7963. Unpruned trees grown to purity on 177,576 rows memorise the training partition, and averaging 100 of them does not reduce variance enough to compensate.

Tuning restores the expected ordering. E3, the same model class with a searched configuration of 191 trees at depth 30, minimum leaf size 11 and no class weighting, gains 0.0526 PR-AUC over E2 and 0.0158 over E1. Boosting (E4) adds 0.0087 more, reaching 0.4295 PR-AUC and 0.8295 ROC-AUC, the best discrimination in the study. Its searched learning rate of 0.029 with 264 boosting rounds and 81 minimum samples per leaf describes a deliberately slow, heavily regularised ensemble — what early stopping selects on a 6.2:1 imbalanced problem.

The support vector machine (E5) is weakest on every discrimination metric (0.3598 PR-AUC, 0.8066 ROC-AUC), for two conflated reasons. Training on 25,000 of 177,576 rows removes 86 % of the data, and Platt scaling on that smaller surface pushes probabilities toward zero, so at threshold 0.5 it detects only 4.6 % of positives. Its tuned-threshold recall recovers to 0.6226, showing the default-threshold behaviour is a calibration artefact rather than a ranking failure. E5 is nonetheless a documented compromise, not a like-for-like competitor.

C. The deep-learning ladder

The four networks land in a tighter band still: 0.4233, 0.4239, 0.4260 and 0.4257 PR-AUC for E6, E7, E8 and E9. Each design change targeted a specific mechanism, and the size of the resulting change is evidence about which mechanisms actually bind here.

Regularisation barely moved discrimination. E7, adding batch normalisation and dropout 0.3 to a network three times the depth of E6, improved PR-AUC by 0.0006, well inside run-to-run noise. It narrowed the train–validation gap, but the gap was not the binding constraint, because the model was already underfitting the signal in twenty-one questionnaire

features. Representation change was more productive: E8, routing the ordinal codes through learned embeddings rather than treating them as real numbers, gained 0.0021 over E7 and became the strongest network. The gain is small but in the direction theory predicts, and it is the only architectural change in the ladder that produced one. Class weighting in E9 left the ranking essentially unchanged, moving PR-AUC by -0.0003, while transforming the operating point: recall at the default threshold rose from 0.153 to 0.792 and MCC from 0.245 to 0.365.0003 while tuned-threshold recall moved by +0.0185.

The comparison that matters most is between paradigms. The best deep model (E8, 0.4260) trails the best classical model (E4, 0.4295) by 0.0035 PR-AUC, while its ROC-AUC is 0.8297 against 0.8295. Neither difference supports a claim of superiority in either direction. On this cohort the flexibility of a neural network buys no measurable discrimination over a well-tuned gradient-boosted ensemble, at an order of magnitude more parameters.

D. Learning curves and the bias–variance diagnosis

Fig. 4 explains the compression of the PR-AUC band rather than merely reporting it, and shows three pathologies. Logistic regression exhibits **high bias**: its training and validation curves converge immediately and stay together, a final gap of 0.0014, both plateauing near 0.42. More data would not help it; only capacity or better features would, and because the curve is flat at the right-hand edge the linear hypothesis class has extracted everything a linear combination of these features can express.

The random forest shows the opposite, **high variance**: training climbs to 0.9228 while validation sits at 0.4153, a gap of 0.5075 that never closes. The forest has ample capacity and spends it fitting noise. Its validation curve is nevertheless still rising by 0.0023 at the largest available training size, so although the model is variance-limited at this sample size, more data would continue to help — the remedy and the diagnosis act on different terms.

Gradient boosting sits between the two, with a gap of 0.1226 and validation still climbing by 0.0072 per doubling of data. It is the only one of the three improving steeply enough at the right-hand edge to be plausibly **data-limited** rather than capacity-limited, consistent with it also being the best-performing model. The network curves in Fig. 5 tell the same story at finer resolution: validation loss turns upward in E6 and E7 while training loss keeps falling, which is overfitting in real time and is why early stopping with restored best weights was applied throughout. In E9 the curves are noisier because inverse-frequency weighting amplifies the gradient contributions of the minority class.

Figure 4 — Learning curves separate the bias-limited model from the variance-limited ones

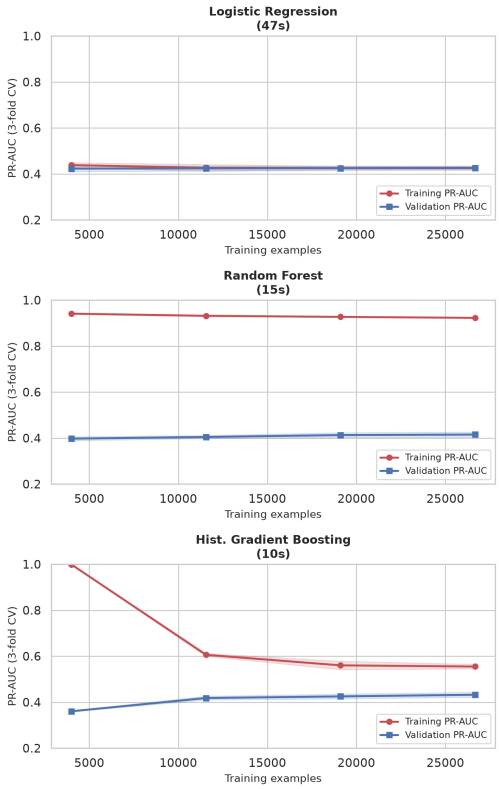


Fig. 4. Learning curves on a 40,000-row stratified subsample, three-fold cross-validation, scored by PR-AUC. Logistic regression shows high bias (gap 0.0014), the random forest high variance (gap 0.5075), and gradient boosting the most favourable trade (gap 0.1226 with validation still climbing).

Figure 5 — Divergence between training and validation loss marks where overfitting begins

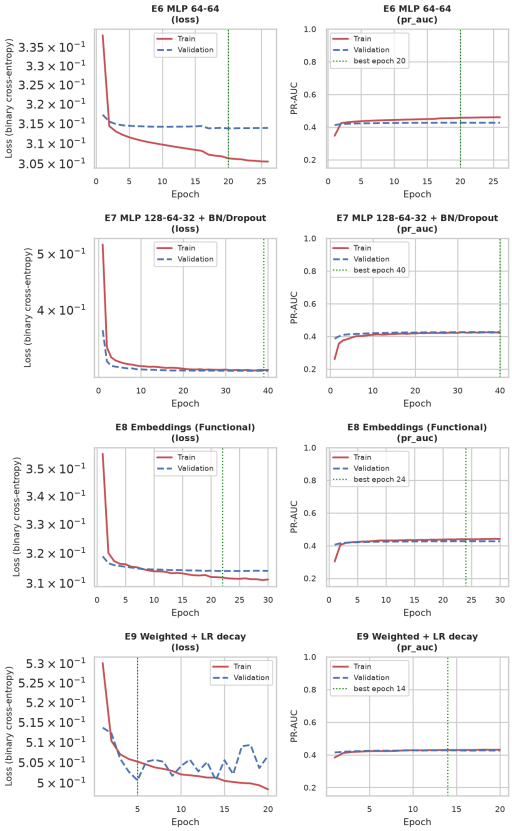


Fig. 5. Training and validation trajectories for the four networks; the loss panels use a logarithmic vertical axis. Validation loss turns upward while training loss keeps falling in E6 and E7, which is overfitting in real time and is why early stopping with restored best weights was applied throughout.

E. Errors, calibration and operating points

Fig. 6 shows where the errors sit. At a fixed 0.5 threshold the strongest model misses 838 of every 1,000 true diabetics while correctly rejecting 98.1 % of non-diabetics — the arithmetic consequence of a 6.2:1 prior combined with an unweighted loss, and the most consequential fact in the results: a screening tool operated at 0.5 would detect one case in six.

Figure 6 — Confusion matrices at threshold 0.5 (row-normalised, raw counts annotated)

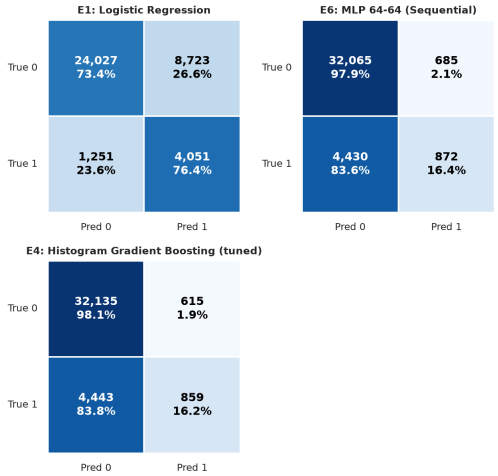


Fig. 6. Row-normalised confusion matrices at threshold 0.5, with raw counts annotated. Each cell reads as the share of respondents truly in that class placed there by the model. At this threshold the strongest model misses 838 of every 1,000 true diabetics.

Fig. 7 explains why ROC-AUC flatters more than PR-AUC. Mean ROC-AUC is around 0.82 while mean PR-AUC is around 0.41; both summarise the same models, but the false-positive rate is deflated by 32,750 true negatives while precision is not. For screening, the PR curve describes the decision actually being made.

Figure 7 — The same models look strong on ROC and merely moderate on precision-recall

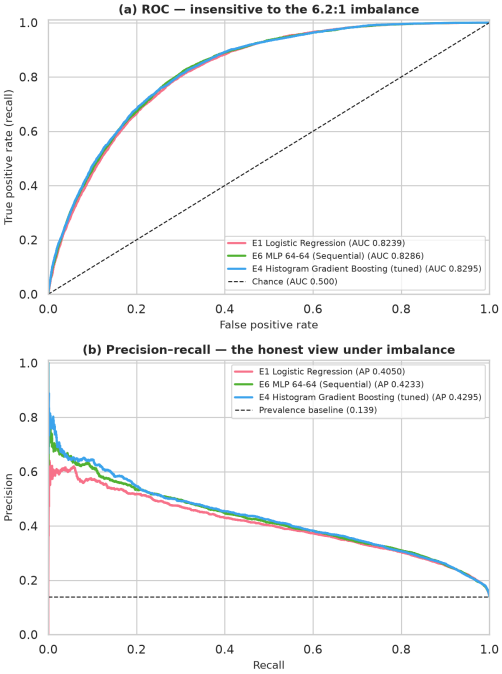


Fig. 7. ROC curves (left) and precision–recall curves (right) for the same four models. The ROC view is flattering because the false-positive rate is deflated by 32,750 true negatives; the precision–recall view is sensitive to exactly the errors that matter for screening.

Figure 8 — Every experiment scores similarly on accuracy, and differently on recall

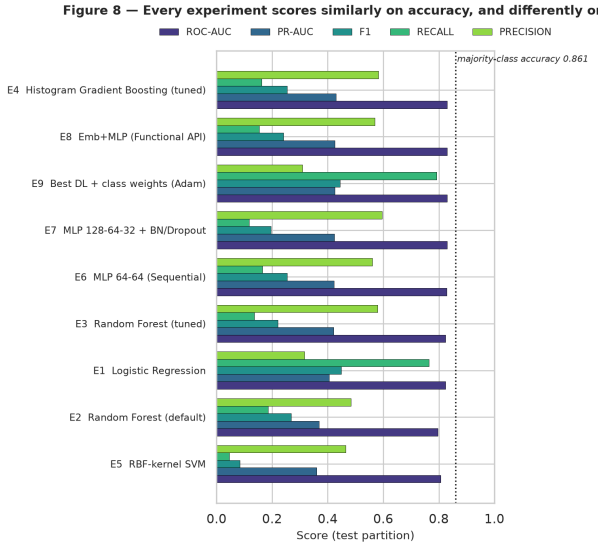


Fig. 8. All nine experiments across five metrics, ordered by PR-AUC. Accuracy varies by less than two points across the whole table while recall varies by 0.72, which is the clearest visual statement of why a single accuracy figure cannot rank these models.

Table III and Fig. 9 make the threshold decision explicit across its range. At 0.5 the best model reaches precision 0.583 with recall 0.162; at 0.3, precision 0.430 with recall 0.481; at 0.2, precision 0.359 with recall 0.677; at 0.1, precision 0.273 with recall 0.874 while flagging 44.7 % of the cohort. The F1-optimal threshold on validation was 0.256, which the test set confirms as near-optimal (tuned F1 0.4654 against the test-optimal 0.4702 at 0.21). Fig. 8 places all nine experiments side by side across five metrics, and moving from 0.5 to 0.256 raises recall by 0.40 — several times the entire spread between the best and worst model in Table II.

Figure 9 — Lowering the threshold trades precision for recall; at 6:1 imbalance that is usually the right trade

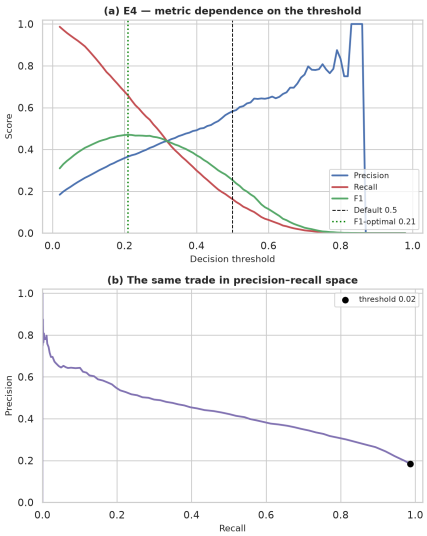


Fig. 9. Threshold behaviour of the strongest model. (a) Precision, recall and F1 as the threshold sweeps; (b) the same trade in precision–recall space. Moving from the default 0.5 to the validation-tuned 0.256 raises recall by 0.40, several times the spread between the best and worst model in Table II.

F. What the ablation shows about feature engineering

Table IV answers the question Section III-D posed. The nine engineered features help the linear model and not the tree models, and the pattern is coherent rather than incidental. Logistic regression gains +0.0086 PR-AUC when they are added, which is what one expects when a linear model is handed interaction terms it cannot construct itself: `AgeBMI` becomes its second-largest standardised coefficient. The random forest moves by -0.0066 and gradient boosting by -0.0020, essentially flat and slightly negative: the signature of features that re-express structure the model already captures while adding noise dimensions to split on. Fig. 10 collects the ablation deltas and the permutation importances.

Standardisation mattered less than expected: the same logistic regression on unscaled features (A7) scored 0.4033 against 0.4050 scaled, a difference of 0.0017, because the solver converges without scaling when all predictors live on comparable bounded scales. The requirement is real for the distance-based SVM but not a general precondition here.

Feature selection was also neutral: removing the five lowest-mutual-information features (A8) cost 0.0010 PR-AUC, confirming that univariate screening cannot identify which features matter in interaction. The duplicate-profile check (A9) is the reassuring result. Refitting A6 on a de-duplicated cohort of 227,908 rows produced 0.4353 PR-AUC against 0.4276 on the full cohort, a gain of 0.0077. Repeated profiles were therefore not inflating the headline figures — if anything they cost a little, because duplicates depress the effective variety of the training signal. The 2.06 % label-noise floor is real but does not explain the ceiling.

TABLE IV. Controlled ablation. Each model family is trained twice with identical hyperparameters, changing only the feature set. A7 isolates standardisation, A8 removes the five lowest-mutual-information features, and A9 refits on a de-duplicated cohort of 227,908 rows.

ID	Configuration	Feat.	ROC	PR-AUC	F1	Rec	t (s)
A1	LR, raw 21 features	21	0.8216	0.3964	0.4438	0.7633	0.1
A2	LR, + 9 engineered	30	0.8239	0.4050	0.4482	0.7641	0.5
A3	RF, raw 21 features	21	0.8204	0.4151	0.2291	0.1435	17.7
A4	RF, + 9 engineered	30	0.8159	0.4085	0.2414	0.1545	25.2
A5	HGB, raw 21 features	21	0.8297	0.4296	0.2563	0.1643	4.3
A6	HGB, + 9 engineered	30	0.8292	0.4276	0.2506	0.1603	4.8
A7	LR, engineered, unscaled	30	0.8233	0.4033	0.4473	0.7618	12.3
A8	HGB, 5 weakest removed	25	0.8282	0.4266	0.2429	0.1541	3.8
A9	HGB, duplicates removed	30	0.8209	0.4353	0.2548	0.1641	4.9

Figure 10 — What the ablation was worth, and what the model actually relies on

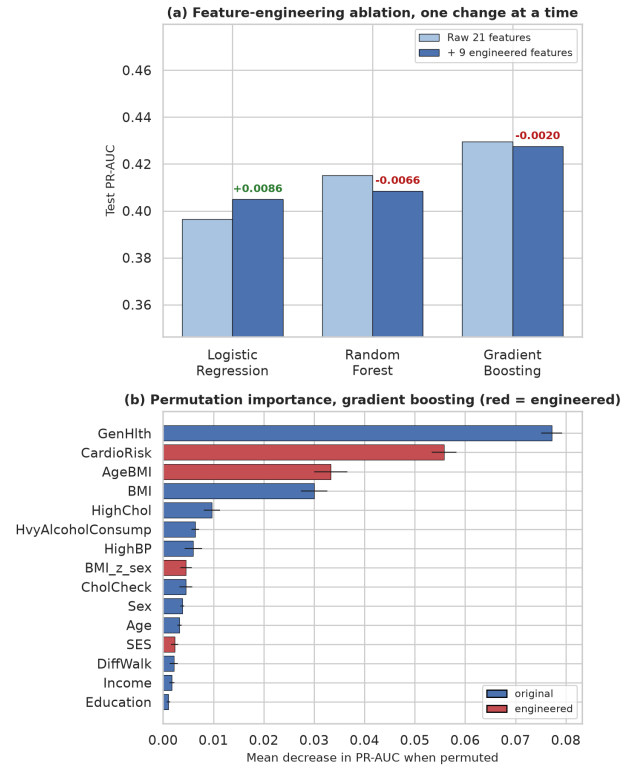


Fig. 10. (a) The feature-engineering ablation for three learners with shared hyperparameters; the engineered features help the linear model (+0.0086 PR-AUC) and not the trees. (b) Permutation importance for the gradient-boosted model, with engineered features in red. Four of the nine engineered features reach the top fifteen.

G. Subgroup error analysis

Recall is strikingly non-uniform, and the pattern is systematic. By self-rated general health it rises monotonically from 0.104 among respondents reporting excellent health to 0.874 among those reporting poor health. By BMI band it rises from 0.196 in the normal-weight band to 0.852 above 40 kg/m². By age it rises from 0.140 in the 30–34 band to 0.669 in the 65–69 band. By income it falls from 0.747 in the lowest band to 0.348 in the highest. Sex makes almost no difference, 0.569 against 0.559.

The false-negative profile explains the mechanism. Compared with correctly detected cases, missed diabetics have a body-mass index 5.7 points lower, self-rated health a full point better and six fewer poor physical-health days. The model is not failing at random; it fails in a specific direction, detecting diabetes well in respondents who already look sick and missing those who are asymptomatic, lean and well — exactly the population a screening programme exists to find. That is a limitation of a symptom-correlated feature set rather than a defect that can be tuned away.

V. Discussion

A. Bias and variance explain the compression

The bias–variance decomposition accounts for the most puzzling feature of the results: nine models of quite different inductive bias produce nearly the same PR-AUC. The noise term is fixed at the 2.06 % label conflict measured in Section III-B plus unmeasured error in self-reported outcomes. The variance term varies enormously, with the random forest's 0.5075 train–validation gap and logistic regression's 0.0014 at opposite extremes. The totals converge because both models fail to add information: the forest's low bias is bought with variance it cannot pay for, and the linear model's low variance is bought with bias it cannot reduce.

Gradient boosting wins because it is the only learner that reduces bias without paying the full variance cost. Shrinkage at 0.029, a floor of 81 samples per leaf and an L2 penalty jointly constrain each tree's contribution, so 264 weak learners combine into a model with both low bias and controlled variance — precisely the mechanism Friedman described [8]. The ablation supports it: once the forest receives comparable regularisation through tuning, its PR-AUC rises from 0.3682 to 0.4208, closing most of the gap. It also explains why the networks could not pull ahead. A network of ample capacity has the same bias-reduction opportunity as boosting, but it was already underfitting, so adding capacity changed nothing.

B. Why the engineered features did not help the trees

The ablation is a null result and it should have been expected. A decision tree can represent any interaction between inputs through successive axis-aligned splits, so `AgeBMI` and `CardioRisk` are already expressible within a single tree of sufficient depth. Adding them does not enlarge the hypothesis class; it re-parameterises it, and the re-parameterisation costs extra search space and a little overfitting. The linear model, by contrast, cannot represent the product of two inputs at all, which is why the same features moved its PR-AUC by +0.0086 and made `AgeBMI` its second-largest coefficient. The practical rule is that interaction-style feature engineering should be targeted at low-capacity models, and that on tree ensembles its expected value is close to zero.

C. The threshold is the decision

The most operationally important finding concerns the decision rule rather than the classifier. Moving from 0.5 to 0.256 raised recall by 0.40, and at 0.1 the best model recalls 87.4 % of cases; no architectural change in this study produced a change of comparable size. The reason is structural: a classifier outputs a ranking, and the threshold converts that ranking into a policy. On imbalanced data the default encodes an implicit assumption that a false positive and a false negative cost the same, which in screening they do not — a missed diabetic progresses toward nephropathy or amputation, while a false alarm costs one confirmatory blood test.

That reasoning implies a threshold well below the F1-optimal one. At 0.1 the model flags 44.7 % of the cohort: in a population with 13.93 % prevalence it would refer roughly three people for every true case, a reasonable price for a cheap follow-up test and an unacceptable one if the confirmatory pathway is scarce. The correct operating point is therefore a function of local capacity rather than of statistics, and the honest contribution of this work is to report the entire precision–recall curve so that the choice can be made with local costs in view.

D. What the errors reveal

The subgroup analysis is where the results are least comfortable. Recall varies from 0.104 to 0.874 across self-rated health, from 0.196 to 0.852 across BMI bands, and from 0.747 down to 0.348 across income bands. These gradients follow prevalence almost exactly: the model is best where the disease is easiest to see and worst where it is rarest. It therefore performs worst on the asymptomatic, normal-weight, higher-income respondents for whom screening has the most value.

Two mechanisms compound this. Height, weight and diabetes status are self-reported, so body-mass index is understated and the label itself carries error; a respondent who does not know they are diabetic is labelled healthy, which depresses recall in exactly the subgroup where the model already struggles. Separately, the access variables are not merely correlates of disease but determinants of the probability of being diagnosed. A model that learns from `CholCheck`, `AnyHealthcare` and `NoDocbcCost` is partly learning who gets tested, and deploying it could widen the disparity it appears to measure.

E. Limitations

Four limitations bound these conclusions. The features are the ceiling: the strongest single predictor has a Spearman correlation of 0.29 with the target, and no architecture can recover the glucose measurement that defines the label. The cohort is a single United States cross-section, so nothing here establishes transfer to Rwanda or to any population of different age structure, diet or diagnostic capacity. The SVM was trained on a 25,000-row subsample and the searches were deliberately small, so the tuned models are lower bounds on what their classes can achieve.

VI. Conclusion

This study asked how much of the diagnostic signal for diabetes is recoverable from questionnaire and examination data, and which paradigm recovers most of it. On 253,680 BRFSS records the first answer is that a substantial and clinically useful signal is present: gradient boosting reaches 0.4295 PR-AUC and 0.8295 ROC-AUC, and at a threshold chosen for screening it recalls 66 % of cases while referring a quarter of the cohort. The second answer is that the paradigm barely matters: the best classical and best deep models differ by 0.0035 PR-AUC, an order of magnitude less than the effect of choosing the threshold.

Three findings generalise beyond this dataset. Comparisons across model families are dominated by configuration and tuning effort rather than inductive bias, so untuned baselines should never be read as evidence about a model class. Interaction-style feature engineering has near-zero expected value on tree ensembles and real value for linear models, so effort should follow model capacity. And on imbalanced screening problems the threshold is a modelling parameter with larger consequences than architecture, so it should be selected on validation data and justified against the cost of a missed case.

Future work follows the limitations. A three-class formulation separating pre-diabetes from diagnosed diabetes would test whether the merged label discards recoverable structure. Recalibration and external validation on an African cohort would establish whether the relationships transfer. Cost-sensitive learning with explicit misclassification costs would formalise the threshold choice. And an audit of the access variables, by excluding them or by reporting performance conditional on them, would test whether the model's dependence on who gets tested can be separated from its dependence on who is sick.

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